

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 6: Lambert Conformal Conic Projection

Syllabus Source: CAE Oxford (pp. 207-248) | Pass Target: $\geq 90\%$ AviationExam

1. THE STANDARD AVIATION CHART ARCHITECTURE

The Lambert Conformal Conic is the primary chart used for aeronautical plotting and mid-latitude IFR airways. It uses a secant cone intersecting the globe at two Standard Parallels (SP1 and SP2):

- Meridians & Parallels: Meridians are straight lines converging at the pole vertex. Parallels are concentric circular arcs centered on the vertex.
- Great Circles on Lambert: GREAT CIRCLES ARE VIRTUALLY STRAIGHT LINES! (Maximum deviation from a straight line over 1,000 NM is negligible).
- Rhumb Lines on Lambert: Rhumb lines are CURVED LINES, concave to the pole of projection.
- Scale Factor (SF): $SF = 1.0$ along the two Standard Parallels. $SF < 1.0$ between the standard parallels (chart is compressed). $SF > 1.0$ outside the standard parallels (chart expands).

2. CONSTANT OF THE CONE & CHART CONVERGENCE

- Constant of the Cone (n): The convergence factor of the cone: $n = \sin(\text{Parallel of Origin}) = \sin(\text{mean latitude of standard parallels})$. E.g. For standard parallels at 30° and 60° , mean lat = 45° , $n = \sin(45^\circ) = 0.7071$.
- Chart Convergence Formula: Chart Convergence (degrees) = Change of Longitude (dlong) $\times$ n = dlong $\times$ $\sin(\text{Parallel of Origin})$

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Straight Line Tracks on a Lambert

Problem: A straight line is drawn on a Lambert chart (cone constant $n = 0.75$) from A (010°W) to B (030°E) in the Northern Hemisphere. Track measured at 010°W is 080° . Find the track angle crossing meridian 030°E at B. Step 1: Chlong = $10^\circ\text{W} + 30^\circ\text{E} = 40^\circ$. Step 2: Chart Convergence = Chlong $\times$ n = $40^\circ \times 0.75 = 30^\circ$. Step 3: In Northern Hemisphere, meridians converge towards the top (pole). - Flying East, the angle between meridian and track increases: - Track at B = Track at A + Convergence = $080^\circ + 30^\circ = 110^\circ$!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Chart Convergence & Great Circle Bearing on Lambert

SCENARIO (Classic ATPL Lambert Chart Drill): A Lambert Conformal Conic chart has standard parallels at 37°N and 65°N, giving a constant of the cone $n = 0.788$: -
 Waypoint A: 50°N 020°W - Waypoint B: 50°N 060°W - A straight line is drawn on the chart from A to B - The measured track angle on the chart at meridian 020°W (Point A) is 260° True QUESTION: What is the chart convergence between A and B, and what will be the measured track angle at Point B (060°W)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Change in Longitude (dlong): - $d\text{long} = 060^\circ\text{W} - 020^\circ\text{W} = 40^\circ$. Step 2: Calculate Chart Convergence using the Constant of the Cone (n): - Formula: Chart Convergence = $d\text{long} \times n$ - Chart Convergence = $40^\circ \times 0.788 = 31.52^\circ$ (~31.5°)! Step 3: Determine Track Direction Evolution from Point A to Point B: - You are flying WESTBOUND in the Northern Hemisphere. - Meridians converge towards the North Pole at the top of the chart. - As you fly West along a straight line on a Lambert chart, the track angle relative to meridians DECREASES: - Track at Point B = Track at Point A - Chart Convergence - Track at Point B = $260.0^\circ - 31.5^\circ = 228.5^\circ$ True! FINAL ANSWER: Chart Convergence = 31.5°. Measured track at Point B = 228.5° True.

[!] EXAM TRAP (AVIATIONEXAM) - Earth Convergence vs Chart Convergence Trap

- Earth Convergence = $d\text{long} \times \sin(\text{lat})$. It varies with every degree of latitude! -
 Chart Convergence on Lambert = $d\text{long} \times n$. It is CONSTANT EVERYWHERE on the chart because 'n' is fixed! - On the parallel where $\sin(\text{lat}) = n$, Earth Convergence equals Chart Convergence.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: On a Lambert Conformal Conic chart, what type of line does a STRAIGHT LINE represent?

- [A]: A rhumb line
- [B]: A GREAT CIRCLE (with negligible deviation)
- [C]: A parallel of latitude
- [D]: An isogonal

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Lambert projection is constructed so that a straight line drawn between any two points closely approximates a Great Circle track.

Q2: A Lambert chart has a constant of the cone $n = 0.75$. Between longitudes 010°E and 030°E, what is the CHART CONVERGENCE?

- [A]: 20°
- [B]: 15° ($d\text{long} = 20^\circ \times 0.75 = 15^\circ$)
- [C]: 7.5°
- [D]: 30°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Chart Convergence = $d\text{long} \times n = (30^\circ - 10^\circ) \times 0.75 = 20^\circ \times 0.75 = 15^\circ$.

Q3: Where on a Lambert Conformal Conic chart is the Scale Factor (SF) LESS THAN 1.0?

- [A]: North of the northern standard parallel
- [B]: BETWEEN THE TWO STANDARD PARALLELS
- [C]: Outside both standard parallels

- [D]: Exactly along the standard parallels

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Because the cone cuts inside the sphere between the two standard parallels, the map features are slightly compressed ($SF < 1.0$). Along the standard parallels, $SF = 1.0$.